A Mini Project Report of

PCC-201-COM : Data Structures

Submitted by

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Under the Guidance of

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[2026-27]



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CERTIFICATE

This is to certify that the "**Mini Project Report**" submitted by **Fartade Pranav Dashrath (62),** on **"Implementation of Student Database Management system [Hashing]" is work done by her and submitted during 2026–2**027 academic year, in partial fulfillment of the requirements for the award of the degree of **Bachelor of engineering** in computer Engineering, at **Bharati Vidyapeeth's College of Engineering, Lavale, Pune**.

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(Guide)

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Head of Computer Engg. Department

ACKNOWLEDGEMENT

It gives immense pleasure to express our gratitude & thanks to all those who have helped in the timely completion of the Mini-Project work. We have a great pleasure in expressing our deep sense of gratitude and indebtedness to **Prof. Tejal. H. Patil** Guide for their invaluable guidance, constant supervision and sustained encouragement throughout the tenure in this Mini Project work.

We would like to express our thanks to **Dr. Lalit. V. Patil**, Head of Department, Department of Computer Engineering Bharati Vidyapeeth's College of Engineering, Lavale, Pune for their constant support and encouragement.

We would like to express our thanks to **Prof. Dr. R. N. Patil** Principal, Bharati Vidyapeeth's College of Engineering, Lavale, Pune for permitting us to take up the project work. We are thankful to our parents and friends for their kind help and support throughout the course.

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Student Name



ABSTRACT

The Flight Reservation System is a Python-based mini project developed to demonstrate the practical application of searching and sorting algorithms in Data Structures. The system manages basic flight information such as flight number, airline name, source, destination, departure time, ticket fare, and seat availability.

The application allows users to view available flights, search for flights using their flight numbers or destinations, sort flights according to ticket fare, and simulate a flight reservation. Linear Search is used to find records in an unsorted list, while Binary Search demonstrates efficient searching in a list sorted by flight number. Bubble Sort is used to arrange flight records according to ticket fare.

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1. INTRODUCTION

1.1 Overview

The Flight Reservation System is a Python-based mini project designed to manage basic flight information and demonstrate the practical implementation of searching and sorting algorithms in Data Structures.

In the modern transportation system, passengers need to search for available flights, compare ticket prices, check departure and destination details, and select suitable flights. Managing this information manually can be time-consuming, especially when multiple flight records are involved.

The proposed Flight Reservation System provides a simple, menu-driven application that allows users to view flight details, search for flights, sort flights according to ticket prices, and simulate the reservation process. The system stores information such as flight number, airline name, source, destination, departure time, ticket fare, and available seats.

1.2 Objectives

The main objectives of the Flight Reservation System are:

1. To develop a simple, menu-driven flight reservation application using Python.
2. To store and manage flight information, including flight number, airline name, source, destination, departure time, ticket fare, and available seats.
3. To implement Linear Search for finding flight records in an unsorted list.
4. To implement Binary Search for searching flight numbers in a sorted list.
5. To implement Bubble Sort to arrange flight records in ascending order of ticket fare.
6. To allow users to display available flights and view their details.
7. To simulate flight reservation by selecting a flight and updating the available seat count.
8. To demonstrate the practical application of searching and sorting algorithms in a real-world-inspired system.
9. To understand the importance of organizing data for efficient retrieval and processing.
10. To provide a simple educational project that helps students understand the role of Data Structures in reservation management systems.

2. LITERATURE SURVEY

Various researchers have studied airline reservation systems to improve booking operations, information management, and customer convenience. Research on sorting algorithms has also explored methods for understanding and applying these algorithms effectively. The following studies provide a foundation for the development of the proposed Flight Reservation System using searching and sorting techniques.

Year	Paper Name	Author	Conclusion
2025	Smart Flight Booking & Management System	Maddipudi Janakiram et al.	The study presents a flight booking and management system with modules for passenger information, flight details, reservations, payment processing, and ticketing. It emphasizes structured database management and data consistency.
2025	Building Scalable Airlines Reservation Systems: A Microservices Approach Using AI and Deep Learning for Enhanced User Experience	Biman Barua, Mohammad Jashim Uddin Mozumder, M. Shamim Kaiser, and Imon Barua	The study proposes a scalable airline reservation architecture using microservices, AI-based personalization, and real-time flight updates to improve system flexibility, responsiveness, and reliability.
2025	Trends and Challenges in AI-Driven Microservices for Cloud-Based Airline Reservation Systems: A Review	Biman Barua, Imon Barua, M. Shamim Kaiser, and Mohammad Jashim Uddin Mozumder	The review examines developments and challenges associated with AI-driven microservices in cloud-based airline reservation systems, providing background on system architecture and scalability.
2025	Real-Time Performance Optimization of Travel Reservation Systems Using AI and Microservices	Biman Barua and M. Shamim Kaiser	The study investigates the use of AI and microservices to improve reservation-system performance, including demand forecasting, resource allocation, scalability, and the handling of high booking traffic.

3. UML DIAGRAM

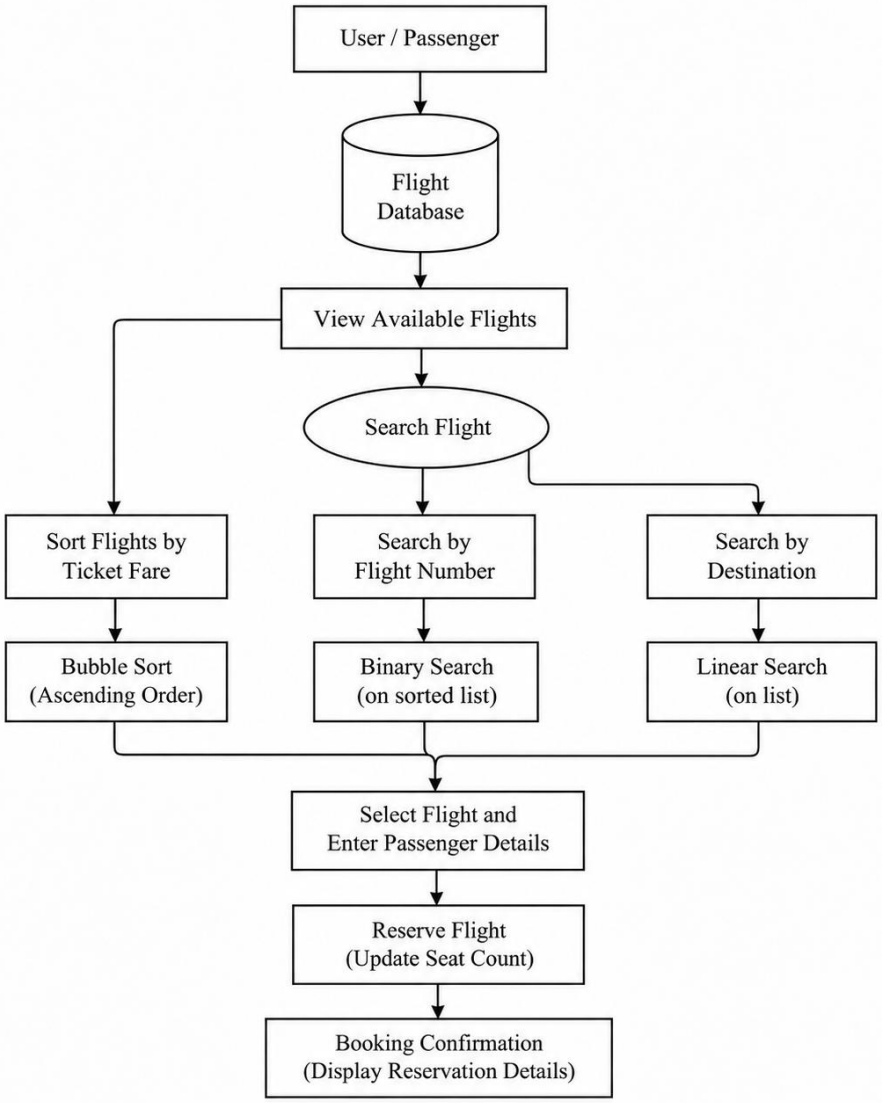


Fig. 3.1: Use Case Diagram of Flight Reservation System

4. SYSTEM REQUIREMENT

4.1 Hardware Requirements

The following hardware is recommended for running the Flight Reservation System:

1. **Operating System:** Windows 10 / Windows 11 or equivalent.
2. **Processor Type:** Intel Core i3 or equivalent processor.
3. **Processor Speed:** 2.0 GHz or higher recommended.
4. **RAM:** Minimum 2 GB; 4 GB recommended.
5. **Hard Disk Space:** At least 100 MB of free storage for the program and project files.
6. **Input Devices:** Keyboard and mouse.
7. **Output Device:** Monitor or display screen.

4.2 Software Requirements

The following software is required to develop and execute the project:

1. **Programming Language:** Python 3.x.
2. **Code Editor / IDE:** IDLE, Visual Studio Code, PyCharm, or Jupyter Notebook.
3. **Operating System:** Windows, Linux, or macOS.
4. **Python Libraries:** Built-in Python features and standard libraries, if required.
5. **Execution Environment:** Python interpreter.
6. **Data Storage:** Python lists and in-memory records for storing flight details during program execution.

5. SYSTEM DESIGN

Existing System

In a traditional flight reservation process, flight information may be maintained in registers, spreadsheets, or basic computerized systems. Finding a particular flight and comparing ticket prices can require additional time and effort when records are not properly organized.

The user may need to check multiple flight records to identify a suitable flight, compare fares, and verify seat availability. Manual searching and sorting become less convenient as the number of flight records increases.

1. **Time-Consuming Search:** Finding a particular flight may require checking multiple records.
2. **Difficulty in Price Comparison:** Flights may not be arranged according to ticket prices.
3. **Unorganized Records:** Flight information may be difficult to maintain and retrieve efficiently.
4. **Manual Processing:** Searching, arranging, and selecting flight records may require additional effort.

Proposed System

The proposed Flight Reservation System is a Python-based, menu-driven application that manages sample flight records and demonstrates the practical use of searching and sorting algorithms.

The system stores flight details such as flight number, airline name, source, destination, departure time, ticket fare, and available seats. Users can view flight records, search for flights, sort flights by ticket fare, and simulate a reservation.

1. **Flight Record Management:** Stores and displays flight details in a structured list.
2. **Flight Search:** Allows users to search for a flight by flight number or destination.
3. **Linear Search:** Checks flight records sequentially to find a matching flight.

Entity Relationship Diagram:

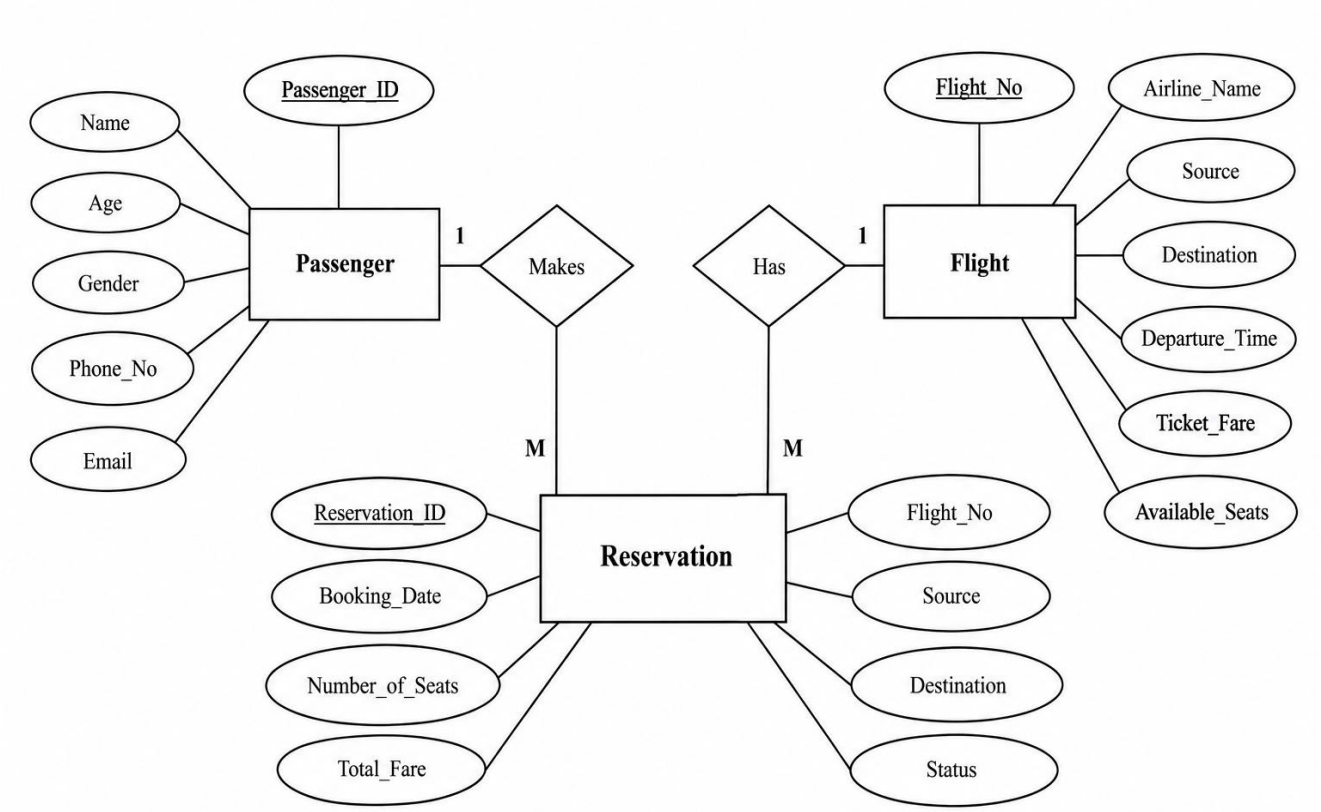
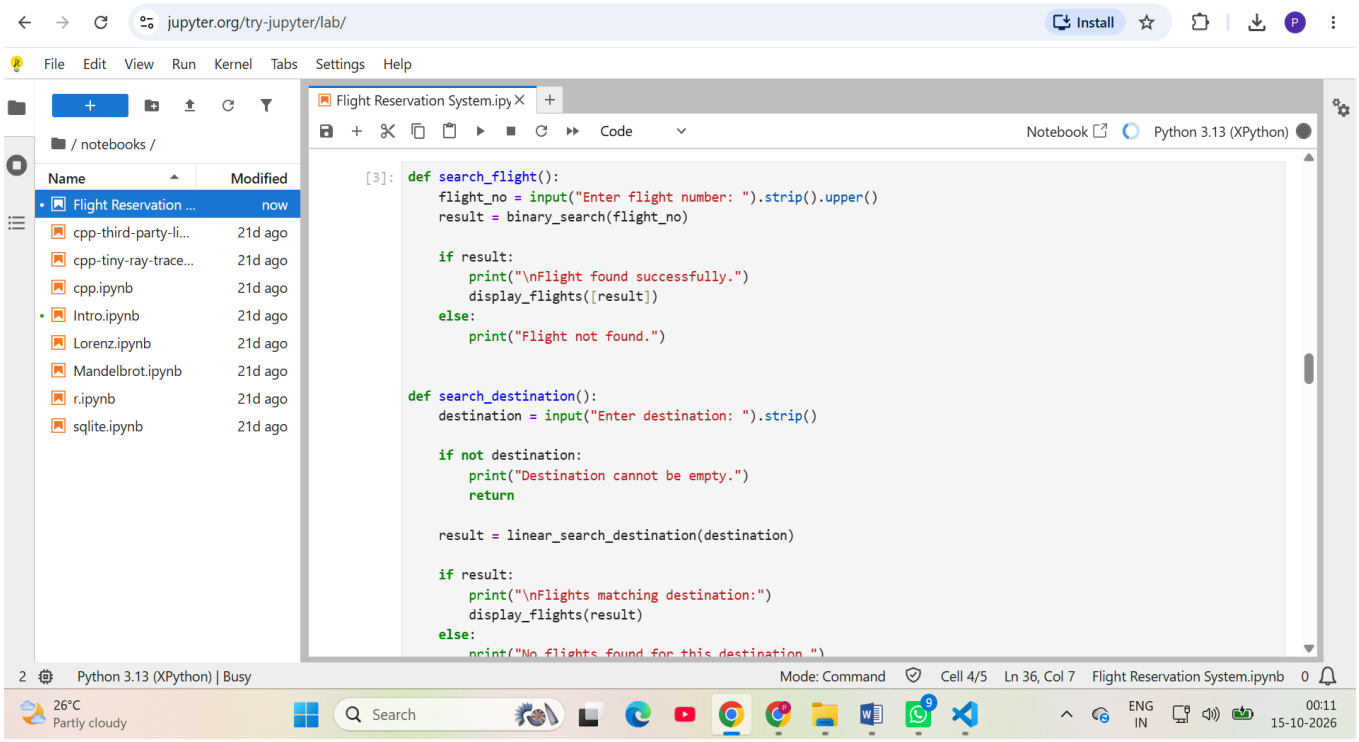


Fig. 5.1: ER Diagram of Flight Reservation System

SYSTEM CODE



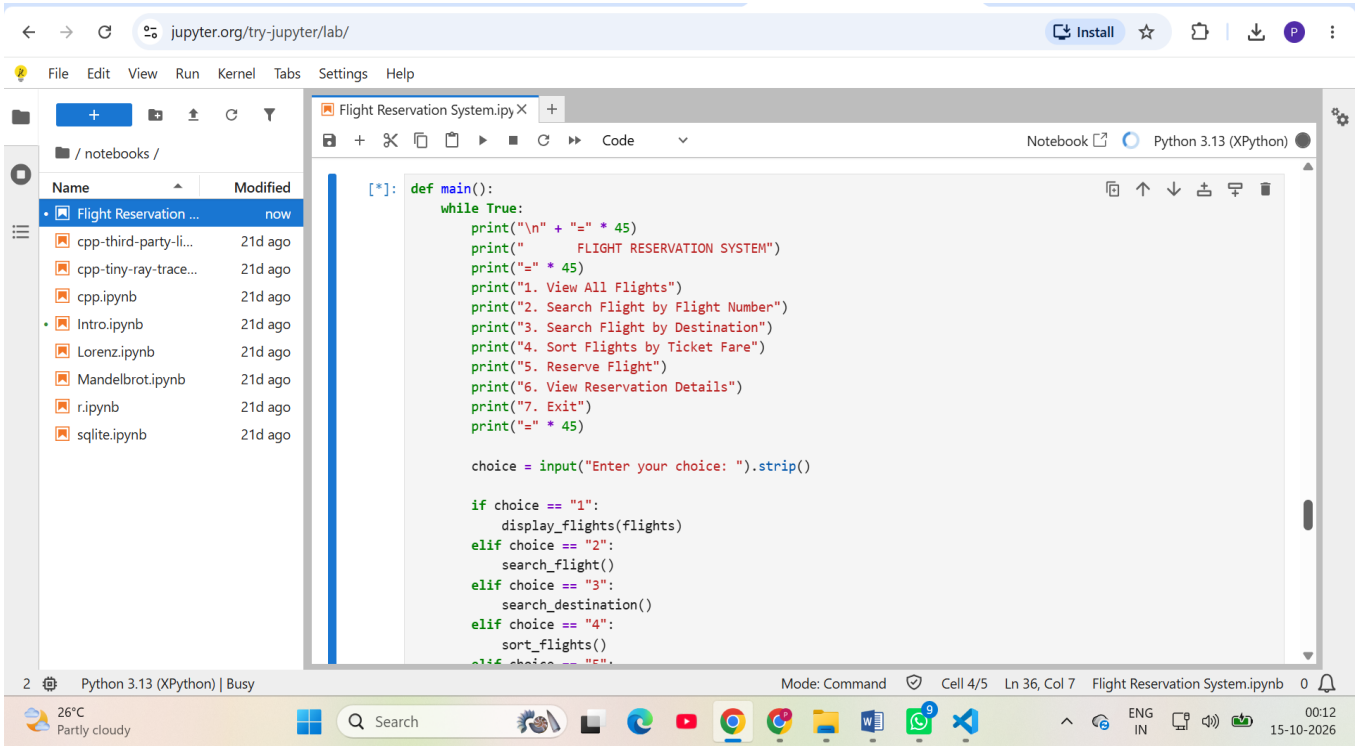
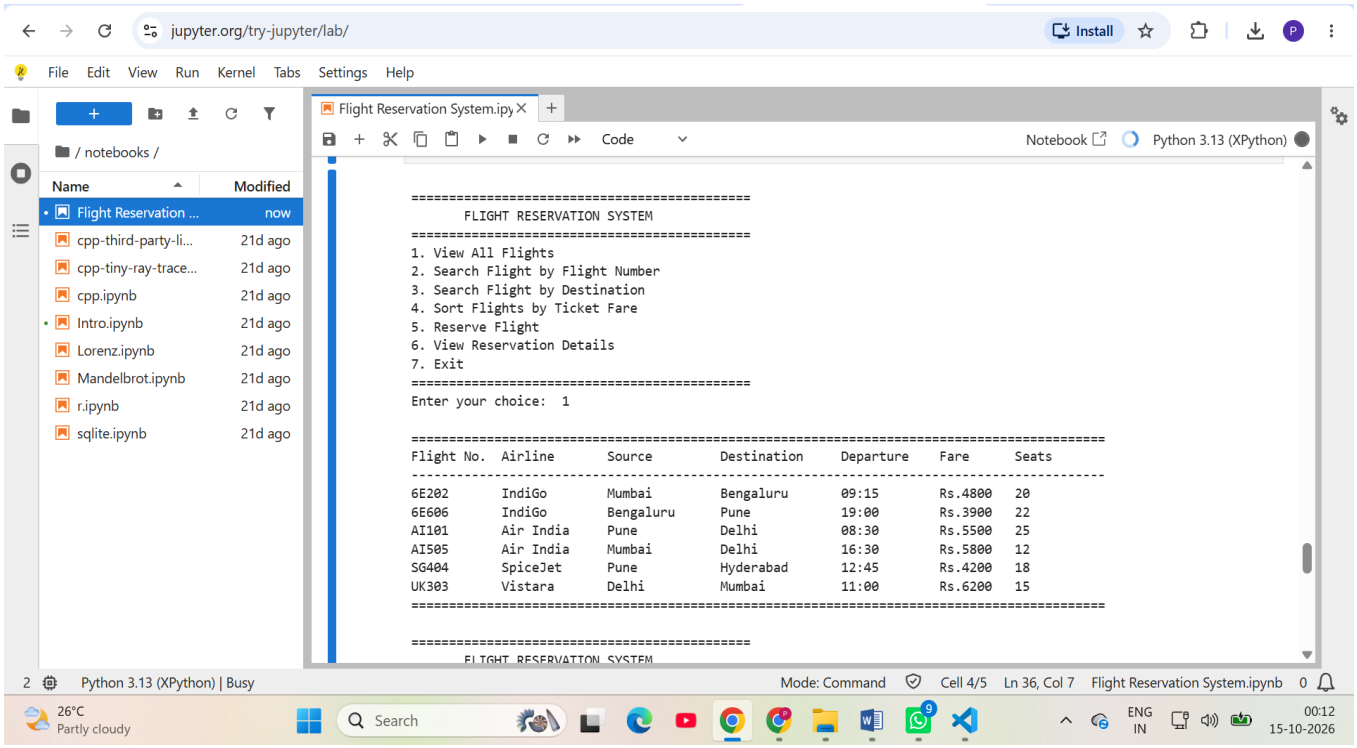
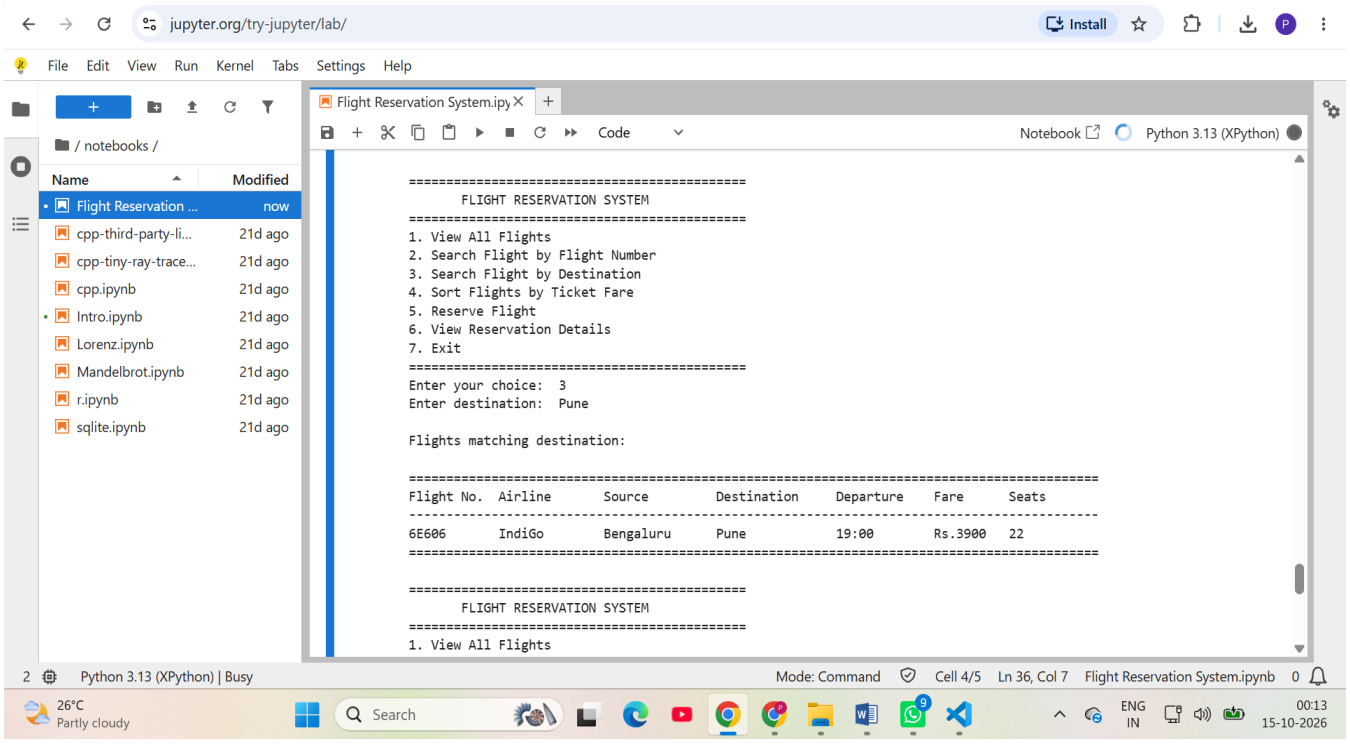


Fig 5.2: System code

6. RESULT


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FLIGHT RESERVATION SYSTEM
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3. Search Flight by Destination
4. Sort Flights by Ticket Fare
5. Reserve Flight
6. View Reservation Details
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6E606 IndiGo Bengaluru Pune 19:00 Rs.3900 22
AI101 Air India Pune Delhi 08:30 Rs.5500 25
AI505 Air India Mumbai Delhi 16:30 Rs.5800 12
SG404 SpiceJet Pune Hyderabad 12:45 Rs.4200 18
UK303 Vistara Delhi Mumbai 11:00 Rs.6200 15
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FLIGHT RESERVATION SYSTEM
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2. Search Flight by Flight Number
3. Search Flight by Destination
4. Sort Flights by Ticket Fare
5. Reserve Flight
6. View Reservation Details
7. Exit
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7. APPLICATIONS

The Flight Reservation System demonstrates the practical application of searching and sorting algorithms in flight information management. It provides a simple way to organize flight records, search for available flights, compare ticket prices, and simulate flight reservations. The concepts used in this project can be applied in various areas of the airline and transportation industry.

One of the major applications of this system is in airline reservation systems. These systems maintain information about flights, departure times, destinations, ticket fares, and seat availability. Searching algorithms help locate specific flight records, while sorting algorithms arrange flights according to ticket prices or other selected criteria.

Another important application is flight search and information retrieval. Users can search for flights using their flight numbers or destinations. Linear Search is useful for finding matching records by checking them sequentially, whereas Binary Search can locate a flight number efficiently when the records are arranged in sorted order.

The system can also be used for ticket price comparison. Passengers often compare ticket prices before selecting a flight. Sorting algorithms can arrange available flights from the lowest fare to the highest fare, making it easier for users to identify affordable options and select flights according to their budgets.

8. FUTURE WORK

The Flight Reservation System can be further improved by adding advanced features to make it more efficient, user-friendly, and reliable. In the future, the system can be connected to a database such as MySQL or SQLite to store flight details, passenger information, and reservation records permanently. This would allow the application to retain records even after the program is closed or restarted.

A graphical user interface (GUI) can be developed using Python libraries such as Tkinter or PyQt to make the system easier to use. Instead of entering commands through a menu-driven console, users could search for flights, compare ticket prices, view seat availability, and make reservations through an interactive interface.

9. CONCLUSION

The Flight Reservation System using Searching and Sorting Algorithms was developed successfully using the Python programming language. The main objective of this project was to demonstrate the practical application of Data Structures and Algorithms in managing flight information and simulating the reservation process. The system allows users to view available flights, search for flights by flight number or destination, sort flights according to ticket fare, and reserve seats by entering passenger details.

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